Objective type questions

8. (iii) trinomial 9. (i) 7 10. (iii) 3

Exercise 3.4

1. $m = 3$ 2. 6 3. 6 4. -1 5. 6 6. 36

Challenge problems

7. 65536 8. $x = 3$ 9. 2 10. 2

11. 21 12. $5x^2 - 11x - 5$; 2 13. $2x^2 - 2xy + 4z^2$; 2

4. Geometry

Exercise 4.1

1. yes 2. cannot draw a triangle

3. (i) 45° (ii) 62° (iii) 30° (iv) 17°
   (v) 18° (vi) 20° (vii) 24° (viii) 27°

4. $\angle A = 60°$; $\angle B = 40°$ 5. 360° 6. 45°, 60°, 75°

7. 55°, 60°, 65° 8. 30°, 60°, 90° 9. $\angle X = 60°$; $\angle Z = 48°$

10. $\angle A = 29°$; $\angle C = 61°$ 11. $\angle M = 38°$; $\angle O = 52°$ 12.(i) 110° (ii) 11°

13. 21° 14. 110° 15. 120°

Objective type questions

16.(ii) 40°, 60°, 80° 17. (iii) 80°, 35° 18. (ii) 68°

19. (iii) $x + y + z = 2(a + b + c)$ 20.(iii) 35° 21.(iii) 130° 22.(ii) 65°, 80°

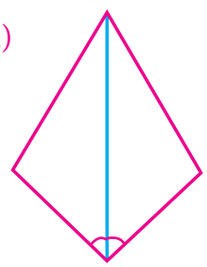
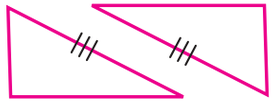
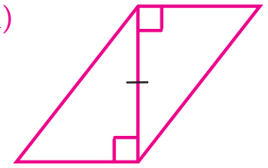
Exercise 4.2

1. Corresponding sides : *AB, DE; BC, EF; AC, DF*
   Corresponding angles : $\angle ABC$, $\angle DEF$; $\angle BCA$, $\angle EFD$; $\angle CAB$, $\angle FDE$

2. (i) $\overline{PQ} = \overline{LN}; \overline{PR} = \overline{LM}; \overline{RQ} = \overline{MN}$
   $\angle RPQ = \angle NLM; \angle PQR = \angle LNM; \angle PRQ = \angle LMN$
   (ii) $\overline{QR} = \overline{LM}; \overline{RP} = \overline{LN}; \overline{PQ} = \overline{MN}$
   $\angle PQR = \angle LMN; \angle QRP = \angle MLN; \angle RPQ = \angle LNM$

3. (i) Not corresponding angles (ii) Not corresponding angles
   (iii) corresponding angles (iv) Not corresponding sides
   (v) corresponding sides (vi) Not corresponding sides

4. (i) congruent triangles by SAS (ii) congruent triangles by SSS
   (iii) congruent triangles by AAA (iv) congruent triangles by RHS
   (v) congruent triangles by SSS (or) RHS (or) SAS

5. (i) 
   (ii) 
   (iii) 

Answers T II EM.indd 104 28-05-2019 19:58:42